

StoreRx

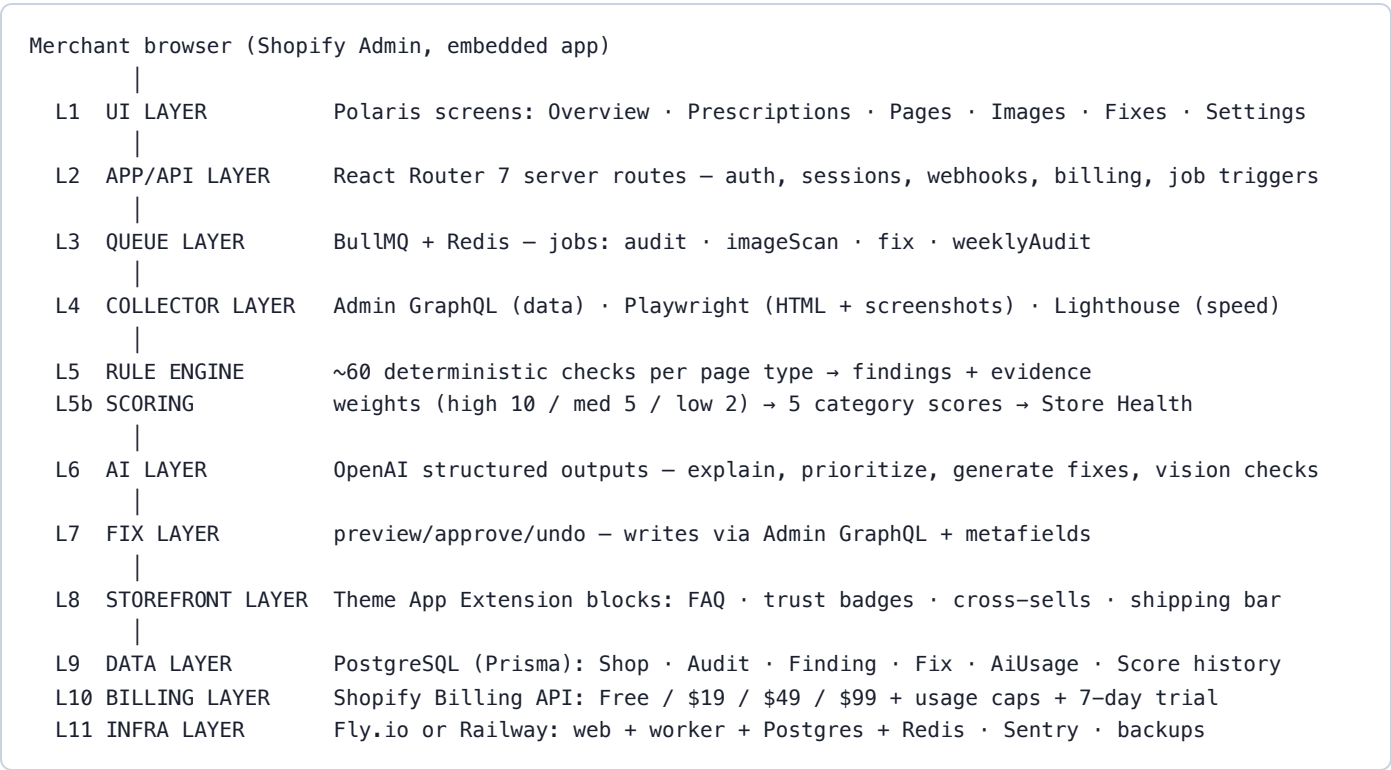
Diagnose. Optimize. Convert. — AI Conversion Doctor for Shopify · Full build plan · September 2026

What it does: A merchant installs the app. StoreRx scans the store page by page (Homepage → Collection → Product → Cart → Checkout), gives scores for Conversion, UX, Performance, SEO and Product Pages, shows a prioritized list of problems ("prescriptions"), and can fix many of them with AI — always with merchant preview and approval, never silently.

1. Golden rules

- **Code finds problems. AI explains and fixes.** Every check is a deterministic rule; the AI never "guesses" whether an issue exists.
- **AI never changes the store without approval.** Every fix: preview → approve → apply → undo (30 days).
- **No invented numbers.** Impact is High / Medium / Low — never "+12% conversion".
- **All AI calls use structured JSON output** through one wrapper function (easy to swap OpenAI ↔ Claude later).
- **Heavy work runs in background workers** — never inside a web request.
- **Storefront changes only via Theme App Extension** — never edit the merchant's theme code.

2. All layers of the system



Layer	Technology	Responsibility
L1 UI	Polaris, App Bridge	Embedded dashboard the merchant sees inside Shopify admin.
L2 App/API	React Router 7 + @shopify/shopify-app-react-router	OAuth, sessions, GDPR webhooks, billing routes, starts jobs. No heavy work here.
L3 Queue	BullMQ + Redis	Runs audits (1–3 min) and fixes in background; retries; progress updates to UI.
L4 Collectors	Admin GraphQL, Playwright, Lighthouse	Gathers facts: products, media, top sellers, page HTML, mobile+desktop screenshots, speed metrics.
L5 Rules + Scoring	TypeScript, unit-tested	Deterministic checks (e.g. "reviews above fold?", "image > 500 KB?"). Produces findings with evidence, then scores.
L6 AI	OpenAI SDK (gpt-4.1-mini / gpt-4.1)	Explains findings in merchant language, ranks priority, writes descriptions / SEO / FAQ / alt text, judges image quality (vision).
L7 Fix	Admin GraphQL mutations	Applies approved fixes; keeps "before" snapshot; undo for 30 days.
L8 Storefront	Theme App Extension (Liquid, <10 KB JS)	Blocks merchant adds in theme editor: FAQ, badges, cross-sells, shipping bar.
L9 Data	PostgreSQL + Prisma	Shops, audits, findings, fixes, AI token usage, score history for trend graphs.
L10 Billing	Shopify Billing API	4 plans, trial, per-plan AI usage caps to protect margins.
L11 Infra	Fly.io / Railway, Sentry	Hosting, scaling, monitoring, logs, backups.

How one audit runs (pipeline)

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Shopify store
├─ Admin GraphQL → products, collections, media, top sellers, theme, apps, shop settings
├─ Playwright    → HTML + screenshots (mobile 390px + desktop 1440px) per page
└─ Lighthouse    → Performance score, LCP, CLS, INP, TBT, script weight
    │
    ▼
Rule engine (rules/<page>.ts) → findings[] {id, severity, evidence, fixableByAI}
    │
    ▼
Scoring → per-page CRO + Perf score → overall Store Health
    │
    ▼
OpenAI (structured JSON) → explanations, priority, recommendations
    │
    ▼
Dashboard → Prescriptions → "Fix with AI" → Preview → Approve → Apply → Undo

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Runs as a background job (1–3 min) with live progress: "Scanning product pages... 3/5". Screenshots and HTML are cached so fixes don't re-crawl.

3. What gets checked (summary)

Page	Main checks
Homepage	Hero CTA above fold (mobile), announcement bar, featured products, trust elements, popup count, navigation, footer links
Collection	Filters/sort, price on cards, quick add-to-cart, thin/empty collections, image consistency, description
Product	Reviews above fold, sticky CTA, price/stock/delivery near CTA, description length, FAQ/size guide, trust badges, cross-sells, out-of-stock variants, SEO title/meta, structured data
Cart	Drawer, free-shipping bar, upsells, trust badges, express checkout buttons
Checkout	(via Admin API) Shop Pay / Apple Pay enabled, guest checkout, shipping & payment options
Images	Size >500 KB, dimensions, missing alt text, mixed ratios, <3 images, duplicates + AI vision: clutter, watermark, blur, off-center
Performance	Lighthouse LCP/CLS/INP per page + which installed apps add how many KB of scripts

Audit samples pages (home, 3 collections, 5 top-selling products, cart) — full image catalog is a separate job on paid plans.

4. AI fixes ("Fix with AI")

Fix	Model	Applied to	Plan
Product description	gpt-4.1	product.descriptionHtml	\$19+
SEO title + meta description	gpt-4.1-mini	product.seo	\$19+
Image alt text	gpt-4.1-mini (vision)	media alt	Free (20) / unlimited \$19+
FAQ section	gpt-4.1-mini	metafield → theme block	\$19+
Trust badges / CTA copy / cross-sells / shipping bar	gpt-4.1-mini	theme extension blocks	\$49+
Image compression / background removal	sharp / image API	re-upload after approval	\$19+ / \$49+

AI token cost per merchant — OpenAI vs Claude

List prices per 1M tokens (input / output), Sept 2026 — verify before launch: OpenAI gpt-4.1 \$2/\$8 · gpt-4.1-mini \$0.40/\$1.60 · Claude Opus 5 \$5/\$25 · Claude Sonnet 5 \$2/\$10 · Claude Haiku 4.5 \$1/\$5.

Task (typical tokens)	OpenAI	Claude
1 audit explanation (~10k in / 2k out)	gpt-4.1-mini ≈ \$0.007	Haiku 4.5 ≈ \$0.02 · Sonnet 5 ≈ \$0.04
1 product description (~3k in / 1k out, best model)	gpt-4.1 ≈ \$0.014	Sonnet 5 ≈ \$0.016 · Opus 5 ≈ \$0.04
1 SEO title + meta (~2k in / 0.2k out)	mini ≈ \$0.001	Haiku ≈ \$0.003
1 image alt text (vision, ~1.5k in)	mini ≈ \$0.001	Haiku ≈ \$0.002
Full vision scan, 200 products × 4 images	≈ \$0.6–1	≈ \$1.5–2

Monthly cost per merchant	OpenAI	Claude	Revenue
\$19 Starter (4 audits, 30 descriptions, 100 alt texts)	≈ \$0.50	≈ \$0.80 (Sonnet+Haiku) / \$1.50 (Opus for copy)	\$19
\$49 Growth (adds full vision scan + more fixes)	≈ \$1.50–2	≈ \$2.50–3.50	\$49

Conclusion: with either provider, AI cost is only **2–7% of plan revenue** — margins are safe. OpenAI (gpt-4.1-mini) is ~40% cheaper overall; Claude (Sonnet 5) writes slightly better long copy and Haiku 4.5 is fine for short tasks. The app uses one generate () wrapper, so you can start with OpenAI and A/B test Claude for descriptions later — or mix both. Add per-plan usage caps either way.

5. Production deployment

Item	Choice	Notes
Hosting	Fly.io or Railway	2 processes: web (app) + worker (audits/fixes). Worker needs 2 GB RAM for Playwright + Lighthouse.
Database	Managed PostgreSQL	Daily backups on. Prisma migrations in deploy step.
Redis	Managed (Upstash / Fly Redis)	Queues + audit progress.
Secrets	Platform env vars	SHOPIFY_API_KEY/SECRET, DATABASE_URL, REDIS_URL, OPENAI_API_KEY, SENTRY_DSN. Never in git.
CI/CD	GitHub Actions	test → build → deploy on push to main; shopify app deploy for extension versions.
Monitoring	Sentry + platform logs	Alert on failed jobs, webhook errors, OpenAI error rate.
Scaling	Queue concurrency	Lighthouse concurrency 1–2 per worker; add workers as installs grow.
Costs (start)	≈ \$25–50/month	Hosting + DB + Redis + Sentry free tier + OpenAI usage.

App Store launch checklist

- GDPR webhooks (customers/data_request, customers/redact, shop/redact) — implemented and tested
- Billing works for all 4 plans + trial; uninstall cleans data within 48 h
- Privacy policy (mentions OpenAI processing, no training on data), support email
- Listing: name check (StoreRx trademark/domain!), icon, 6 screenshots, demo video
- Review wait: 1–3 weeks, expect 1–2 rounds of feedback

6. Pricing

Free	\$19/mo Starter	\$49/mo Growth	\$99/mo Pro
1 audit/month, 5 AI recommendations, 20 alt texts	Weekly audits, unlimited recommendations, product copy fixes, image compression	Unlimited audits, all AI fixes, full vision scan, background removal, conversion analytics	Everything + multi-store, advanced analytics, priority support

7. Timeline — full-time plan (6 h/day, 6 days/week)

Stage 1 — MVP build: ~1 month (26 working days, ≈ 120 hours)

Days	Work	Hours
Day 1–3	Setup: auth, dev store link, database models (Shop, Audit, Finding, Fix, AiUsage), billing 4 plans + trial	15
Day 4–8	Collectors: Admin GraphQL, Playwright crawl + screenshots, Lighthouse, BullMQ queue	25
Day 9–12	Rule engine: 12 product rules + image checks + scoring + fixture tests	18
Day 13–15	AI layer: generate() wrapper, structured outputs, prompts (explain, description, SEO, alt text), token logging	12
Day 16–19	Fix workflow: preview / approve / apply / undo, bulk alt text	18
Day 20–24	Dashboard UI (Polaris): Overview, Prescriptions, Fixes, Settings	22
Day 25–26	Weekly scheduled audits + email summary	8
Total	Working MVP on a dev store	≈ 118 h

Stage 2 — Production hardening: 15–20 days (≈ 90–120 hours)

Days	Work	Hours
Day 1–3	Hosting: Fly.io/Railway web + worker, managed Postgres + Redis, secrets, deploys	15
Day 4–6	CI/CD (GitHub Actions), Sentry monitoring, logging, backups	15
Day 7–11	Real-theme edge cases: password pages, redirects, broken themes, slow stores, retries	28
Day 12–14	GDPR webhooks, uninstall cleanup, billing edge cases (upgrade/downgrade/decline)	16
Day 15–18	Performance + cost hardening: queue concurrency, AI usage caps, rate limits, security pass	20
Day 19–20	Homepage + collection + cart rules (expand beyond product pages)	12

Stage 3 — Testing & launch: 6–10 days (≈ 40–60 hours)

Days	Work	Hours
Day 1–4	Beta test on 3–5 real stores, fix bugs found, polish copy and empty states	25
Day 5–6	App Store listing: name check, icon, 6 screenshots, demo video, privacy policy, support email	12
Day 7–10	Submit for review + answer reviewer feedback (review wait itself is 1–3 weeks — keep improving during it)	15

Total: ~2 months full-time from first line of code to App Store submission (≈ 250–300 hours). The Shopify review wait (1–3 weeks) runs after Stage 3 — use that time for the vision image scan, theme blocks, and trend graphs (+60–80 h post-launch features).